Kubernetes IPAM Efficiency & CNI Resilience

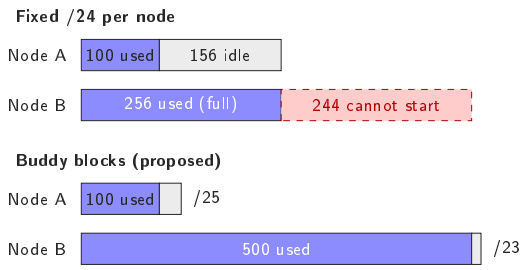
CS331 - Computer Networks, Project 2

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Problem 1: Static Pod IP Allocation

Kubernetes gives every node a **fixed block** of pod addresses (default /24 = 256), once, at creation.

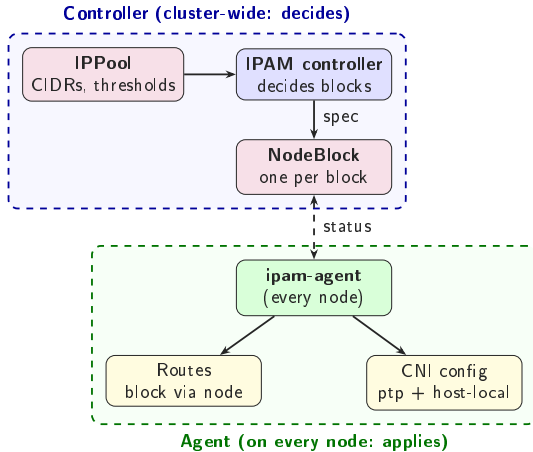
- Quiet nodes waste most of their block
- Busy nodes run out: pods stuck in `ContainerCreating`
- The scheduler does not track IPs
- No resizing, no growth into a new CIDR, no shrinking



Solution 1: Buddy-Style Dynamic IPAM

- Pool = **buddy tree**: /22 → two /23 → . . . → /28
- Each node gets **aligned power-of-two blocks**
  ⇒ one route per block
- **Grow** at 80% use, or at once on "no IP addresses available"
- Prefer the node's **own buddy** ⇒ halves merge
- **Shrink** by draining: no new pods, release when empty; pods never moved
- Spill into **discontiguous** CIDRs, reclaim leaked leases

Block life: Pending → Active → Draining → Quarantined → Released

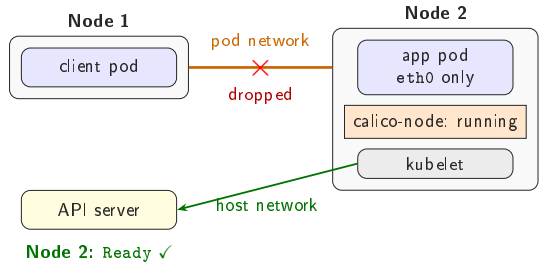


Problem 2: The CNI Is a Single Point of Failure

Every pod is **single-homed**: one interface (`eth0`), one IP, one path, from one CNI.

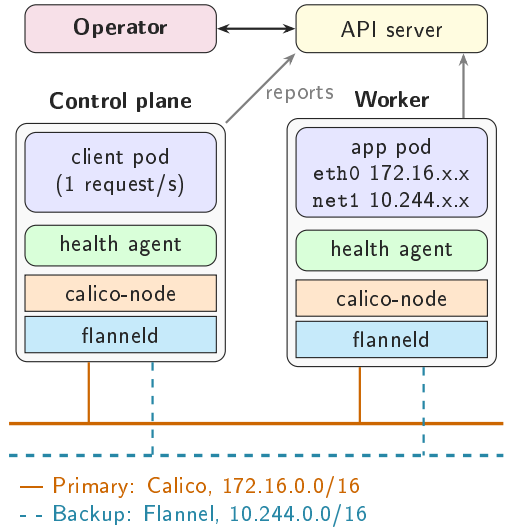
- CNI data path breaks (routes, tunnel, rules) → pod unreachable
- Kubelet uses the **host network**, so the node still looks `Ready`
- Pods are not moved; restarting containers does not fix the network
- fix: manual CNI restart or node reboot

No second path, and no way to detect or repair the failure through the network itself.



Solution 2: Dual-CNI Failover

1. **Two paths**: `eth0` on Calico, `net1` on Flannel (Multus)
2. **Detect**: health agents probe one peer per round over both paths, 3 results in a row to flip
3. **Blame**: a node is broken only if probes in *and* out fail while another pair works, else wait
4. **Fail over**: readiness gate off (`eth0` leaves Service) + **shadow EndpointSlice** with `net1`
5. **Recover**: taint node, restart `calico-node` once
6. **Fail back**: gate on, then delete shadow



Proposed Solution

IPAM efficiency

- Buddy blocks sized to demand
- Grow before a node runs out
- Shrink by draining, pods never moved

CNI resilience

- Backup path on a second CNI
- Fail over only on proof
- Fail back after repair

Thank You!

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